Fiscal Year:	FY 2023	**Task Last Updated:**	FY 08/01/2022
PI Name:	Sanford, Larry Ph.D.		
Project Title:	Impact of Inflight Stress and Sleep Disturbances on Brain Function, Neural Communication, and Inflammation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per L. Juliette (Ed., 5/8/22)		
Key Personnel Changes/Previous PI:	July 2021 report: Shao-Hui Chuang and Richard Ciavarra are no longer working on the project. July 2020 report: None.		
COI Name (Institution):	Britten, Richard Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
Grant/Contract No.:	80NSSC19K1582		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts may experience several physical and psychological challenges including prolonged exposure to microgravity, potential social isolation (SI), and predicted exposure to ~13 cGy/Yr Space Radiation (SR). Each of these stressors can have deleterious effects on physical and mental health as well as negatively impact mission performance. Each also can disturb sleep, which can further exacerbate the effects of stress. Determining how astronauts respond to the single and compound effects of mission related stressors is crucial for understanding and mitigating factors that could impede mission success. However, examining their effects will be complex because they have different time courses and durations. SR exposure will be episodic whereas microgravity will be constant and SI and sleep disturbances can be temporary or prolonged. Their effects may also be additive or synergistic, and may vary with sex and with individual differences in stress resilience and vulnerability. Therefore, our goal in this project is to determine the single and synergistic effects of inflight stress and SR on mission related biological and functional challenges that astronauts may face. We will determine how order of experience impacts the effects of inflight stressors and determine sex and individual differences in stress resiliency using an established Wistar rat model. We will use established ground-based models of microgravity (hindlimb unloading), SI, and SR. We will use telemetry to record sleep and to assess temperature increases as an index of the stress response. We will use plasma levels of brain derived neurotropic factor to predict individual differences in stress responses. We will assess fear learning and extinction to determine how chronic inflight stressors impact ability to respond to an additional acute stressor. We will analyze food entrained, anticipatory activity to assess potential impairments in temporal perception. We will determine the effects of inflight stress and SR on sensorimotor function and recognition memory using behavioral tests. We will use local field potential recordings in the amygdala, medial prefrontal cortex, and hippocampus to determine whether stressors and SR alter the coordinated neural activity underlying fear learning and extinction, and we will determine whether there are structural changes within connections between regions. We will use chemokine/cytokine arrays and Ingenuity Pathway Analysis to determine the effect of stressors and SR on the immune system. Collectively these studies will assess functional, mechanistic, and immune consequences of several stressors inherent in space exploration. The data they provide will aid in risk assessment, and may help to develop countermeasures based on individual differences in ability to withstand stress and/or targeting immune pathways that may confer protection against the deleterious effects of multiple stressors that astronauts will face.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will increase understanding of how individual differences in resilience and vulnerability factor in determining lasting outcomes of stressful experiences. They will also determine the role that stress-induced alterations in sleep play in mediating the outcomes of stress. The results will have importance for understanding how individuals cope with stress on Earth and on missions in space.
Task Progress:	Our goal in this project is to determine the impact and potential synergistic effects that multiple inflight stressors and exposure to space radiation (SR) can have on biological systems and mission related performance. Outcomes that we are examining include sensorimotor function and balance (adhesive removal task and balance beam performance), recognition memory (novel object recognition task), time perception (anticipatory activity), fear memory, and fear memory related neural communication. We are also examining the relationship of altered performance to the effects of inflight stressors on sleep, and how responses to stressors vary in resilient (Res) animals that maintain sleep after experiencing stress and vulnerable (Vul) animals that have reduced sleep after experiencing stress. This is the third year of this project. We have now conducted studies in one cohort of control rats (n=20), one cohort of social isolation (SI) rats (n=21), and one cohort of SR rats (n=30 irradiated (15 cGy GCRsim) and 10 shams). We thus have been able to begin making comparisons that include SI and SR animals, with (n=15) and without (n=15) SI, and also to begin delineating Res and Vul rats within treatment groups. Rats were divided into Res and Vul phenotypes based on rapid eye movement (REM) sleep responses to footshock stress. Some analyses and comparisons across groups are still ongoing. We also have our first group of irradiated female rats. Thirty female rats were irradiated (15 cGy GCRsim) and 15 sham control rats received identical handling, but were not irradiated. The first cohort of female rats has been implanted with telemetry transmitters for recording sleep and has been scheduled for behavioral testing and sleep recording that will be conducted over the next several months. We have made good progress over the last year, especially given the initial impediments to this project imposed by COVID-19 and the budgetary constraints that have limited the pace of our studies. The data we have acquired demonstrates both deficits associated with SR and unexpected effects of SI and SI + SR interactions. Our studies also have found that Res and Vul rats maintain their phenotypes under SI and SR and that Vul rats show greater deficits on some measures. On the Balance Beam task, we found minimal differences between each treatment group's ability to complete the task when the Res and Vul phenotypes were not considered separately. However, the irradiated groups (SR and SI + SR) had fewer individuals that learned the task and deficits were more pronounced in the Vul rats. Similar effects were found on the adhesive removal task test: minimal differences in performance in the task between treatment groups were observed when stress responsive Res and Vul phenotypes were not considered separately. When separated into phenotypes, on average, Vul control and SR rats took longer to notice and remove the adhesive sticker compared to Res rats within their respective treatment group. Because we found that standard behavioral measures on the Balance Beam and adhesive removal task tests were inadequate to fully evaluate behavior, we developed additional assessments of the animals' behavioral repertoires (e.g., exploration, disequilibrium, fear and anxiety, and maintenance (grooming), and irritated behaviors). These assessments demonstrate both motor deficits/changes and interactions between motor behavior and emotion in the treatment groups, and also suggest that altered motor responses do not provide the entire explanation for behavioral performance effects. On the novel object recognition task, Res animals in the SI and SI + SR groups spent more time around the novel object than did Vul animals. Additionally, regardless of phenotype, SR animals, whether alone or in combination with SI, spent less time around the novel object compared to animals treated with SI alone. Our sleep and electroencephalogram (EEG) data analyses show significant alterations in both sleep amounts and EEG spectra in rats subjected to SI and SR. These data demonstrate that SI, SR, and SI + SR can significantly impact the effects of additional stress (produced by footshock training and fear conditioning) on non-REM and REM sleep amounts and can alter EEG spectra in wakefulness and both sleep states. SR resulted in an increase in REM sleep amounts that

	occurred in association with decreased REM theta amplitude. Increased total REM sleep duration, along with decreased REM theta amplitude, is also found in association with chronic social defeat stress in rats. This suggests that the increase in REM sleep amount and decrease in REM theta in SR treated rats may reflect continuing stress or deficits in the ability to cope with stress. We will further explore this possibility in our full studies. One of the most prominent findings in this study is an increase in freezing found in SR and SI+SR rats. Alterations in freezing associated with irradiation, including increases, have been previously reported, though the level of freezing is exceedingly high in our paradigm. This likely relates to the more extensive training regimen (20 footshocks) that we employ compared to other studies. Greater freezing has generally been interpreted as greater fear and stronger fear memory retention. However, we have found no particular correlation between freezing and a number of fear related measures (e.g., neural activity in brain stress regulatory regions, corticosterone, and stress-induced hyperthermia, sleep) indicating that it is not predictive of specific neural changes, and likely is not sufficient as a measure of the type of fear that is being experienced. Additional work will be required to determine whether and how SR differentially impacts neural activity in fear conditioned Res and Vul rats, and its functional relevance. In summary, our data demonstrate both single and synergistic effects of SI and SR on several behaviors and sleep, and also demonstrate differential effects in male Res and Vul rats. These findings suggest that this model may be useful for determining the role of individual differences in the effects of SR and SI in ways that may lead to a better understanding of how stress and SR may differentially impact the health and performance of individual astronauts. Also, as indicated above, we are beginning our first studies with females. Once completed, we can begin making comparisons of the effects of SI and SR, and their interactions, in males and females.
Bibliography Type:	Description: (Last Updated: 03/16/2026)
Abstracts for Journals and Proceedings	Sanford LD, Wellman, LL, Adkins, AM, Hildinger, EM, Britten, RA. "Differential effects of inflight stressors on sleep and behavior in stress vulnerable and resilient rats." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Adkins AM, Pham V Xia, H, Hildinger EM, Wang B, Yoon H, Osher S, Britten RA, Wellman LL, Sanford LD. "Effects of inflight stressors on neural networks regulating emotion and cognition. " 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Poster Session B: Behavioral Medicine (BMed). Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Articles in Other Journals or Periodicals	Britten RA, Sanford LD, Guo M-L, Krishnan B, Emmett MR, Laiakis EC. "A Sankofian appraisal on how to maximize translatability of rodent space radiation/CNS studies to astronauts. " THREE: The Health Risks of Extraterrestrial Environments. 2022 Apr 25. https://three.jsc.nasa.gov/Encyclopedia/Article/436 , Apr-2022
Books/Book Chapters	Wellman LL, Adkins AM, Yoon H, Britten RA, Sanford LD. "Telemetry in rats and mice: Methodological considerations and example studies of stress and anxiety in ground-based spaceflight analogs. " in "Psychiatric Vulnerability, Mood, and Anxiety Disorders: Tests and Models in Mice and Rats." Ed. J. Harro. Neuromethods, Vol. 190. New York, NY: Humana, 2022. p. 201-22. https://doi.org/10.1007/978-1-0716-2748-8_11 , Oct-2022